

Thyroid

Hypothyroidism, hyperthyroidism and Graves disease, TSH and T4 genetics, nodules and goiter, and thyroid cancer

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Thyroid autoimmunity is on the low side. Graves disease is at percentile 3.3 in the genomic score (and in the 18.8 in the 112 variants). The loci loads of hypothyroidism and Hashimoto (56.2) and hyperthyroidism and Graves (50.1) are typical. The major risk alleles are missing: **PTPN22 R620W G/G**, without the haplotype **HLA-DR3/DQ2.5** (rs2187668 C/C), **SH2B3 C/C**, VAV3 and FLT3 without risk alleles. Hypothyroidism scores (42.2 and 38.2) and Hashimoto scores (33.8 and 51.8) are typical. The local conversion of T4 to T3 is preserved (**DIO2 Thr92Ala T/T**, without the Ala92 allele), and **CHEK2 I157T** is absent.

Unfavourable. The structural side of the gland is the weak point. The nodular goiter is at the 90.7 and 93.2 (genomic) percentiles. Thyroid cancer is above the 79th percentile in all scores: 79.1, 82.5, 95.9 (41% coverage) and 80.7 (genomic). The loci load of thyroid cancer is at 93.7. The risk alleles **NRG1 G/G** and **TERT T/T** (2 copies each) and 1 copy in **FOXE1 (A/G)**, **NKX2-1 (C/T)**, DIRC3 and PCNXL2. There is also a slight 1 risk allele in **CTLA4 CT60**, **TPO** and **TSHR** (rs179247).

What does it mean? The genetic thermostat tends to a low TSH (score at percentile 19.1, loci load at percentile 3.9) with a high free T4 (score at percentile 80.1, loci in the 81.7), as already seen in the §43 and the §86. A TSH in the lower half of the reference range may be normal for Sérgio F and, if the T4 and the T3 are free, it is not hyperthyroidism. Genetics makes autoimmune thyroid disease unlikely and puts nodules and differentiated thyroid cancer approximately in the upper quintile. Differentiated thyroid cancer is, in most cases, slow growing and with excellent prognosis. This confirms the §94 (percentile 79.1 thyroid cancer score and high loci load).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 23 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 23 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 15 PGS Catalog scores calculated locally with 5,323,160 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2677 loci, 2337 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the scores of disease, a higher percentile = more risk. In the two levels scores (TSH and T4 free), the percentile indicates only the expected genetic level, without judgment. The scores marked as already used keep the values of previous reports. The Hashimoto of 55 variants was calculated with kNN inference in §18 and is here recalculated with the imputation; the value of the imputation replaces the previous.

Thyroid thermostat

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
TSH (thyroid stimulating hormone)	PGS004906 (Mulder 2023, 60 variants)	57 60 (weighted coverage 95%)	19.1	LOW	without judgment
Free T4	PGS004907 (Mulder 2023, 31 variants)	29 31 (weighted coverage 90%)	80.1	HIGH	without judgment

- **TSH (thyroid stimulating hormone):** In §43 (Hormones) and in §86.

Hypothyroidism and Hashimoto

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hypothyroidism	PGS000965 (Tanigawa 2022, 4 535 variants)	4,043 4,524 (weighted coverage 92%)	42.2	TYPICAL	typical
Hypothyroidism (genomic)	PGS005269 (White 2025, PRS-CS, 1.09 million variants)	1,063,714 1,085,170 (weighted coverage 98%)	38.2	TYPICAL	typical
Lymphocytic thyroiditis (Hashimoto)	PGS005270 (White 2025, 55 variants)	50 55 (weighted coverage 70%)	33.8	TYPICAL	typical
Lymphocytic thyroiditis (Hashimoto, genomic)	PGS005272 (White 2025, PRS-CS, 1.09 million variants)	1,063,703 1,085,142 (weighted coverage 97%)	51.8	TYPICAL	typical

- **Hypothyroidism:** In §43 (Hormones).
- **Lymphocytic thyroiditis (Hashimoto):** In §18 (Inflammation and autoimmunity), there with kNN inference; here with the imputation.

Hyperthyroidism and Graves

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hyperthyroidism (thyrotoxicosis)	PGS001042 (Tanigawa 2022, 226 variants)	198 221 (weighted coverage 91%)	55.5	TYPICAL	typical
Graves' disease	PGS005264 (White 2025, 112 variants)	81 112 (weighted coverage 33%)	18.8	LESS LIKELY	favourable
Graves disease (genomic)	PGS005266 (White 2025, PRS-CS, 1.09 million variants)	1,063,714 1,085,170 (weighted coverage 97%)	3.3	LESS LIKELY	favourable

- **Hyperthyroidism (thyroidosis):** In §43 (Hormones).

Nodules and goiter

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Benign goiter	PGS005261 (White 2025, 110 variants)	99 110 (weighted coverage 51%)	90.7	MORE LIKELY	unfavourable
Benign goiter (genomic)	PGS005263 (White 2025, PRS-CS, 1.09 million variants)	1,063,714 1,085,170 (weighted coverage 97%)	93.2	MORE LIKELY	unfavourable

Thyroid cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Thyroid cancer	PGS000087 (Graff 2021, 12 variants)	12 12 (weighted coverage 100%)	79.1	TYPICAL	typical
Thyroid cancer	PGS005210 (Gibson 2024, 12 variants)	12 12 (weighted coverage 100%)	82.5	MORE LIKELY	unfavourable

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Thyroid cancer	PGS004954 (Pozdeyev 2024, 26 variants)	20 26 (weighted coverage 41%)	95.9	MORE LIKELY	unfavourable
Thyroid carcinoma (genomic)	PGS005260 (White 2025, PRS-CS, 1.09 million variants)	1,063,714 1,085,170 (weighted coverage 97%)	80.7	MORE LIKELY	unfavourable

- **Thyroid cancer:** In §24 (Canker Risk) and in §94 (Radiation).

4. Key variants by topic

Variants of pituitary-thyroid thermostat, thyroid autoimmunity and papillary cancer replicated loci. All were read directly on the chip. The loci of TSH only inform the expected level of the hormone.

Thyroid thermostat (TSH, T4 and conversion to T3)

Genes that fix the equilibrium point of TSH and T4 and the conversion of T4 into T3 in tissues.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
DIO2	rs225014	Thr92Ala	T/T	chip	C (0 copy/s)	favourable	Thr92Ala: C allele (Ala92) reduces local conversion of T4 into T3 in tissues (brain, muscle)	Moderate
DIO1	rs2235544		C/A	chip	C (1 copy/s)	without judgment	C allele increases the activity of 1 type deiodinase: plus free T3 and minus free T4 and reverse T3	Moderate
PDE8B	rs4704397		G/G	chip	A (0 copy/s)	without judgment	Allele A rises to TSH (~0.1 mUI/L per copy); the highest locus of TSH	Strong
CAPZB	rs10917469		A/A	chip	A (2 copy/s)	without judgment	Locus CAPZB of TSH [GWAS Catalog: Thyroid stimulating hormone levels, allele A, beta 0.110 [0.1-0.12] unit increase, p=3E-105, PMID 39067062]	Moderate
VEGFA	rs9472138		C/T	chip	T (1 copy/s)	without judgment	Locus VEGFA of TSH [GWAS Catalog: Thyroid hormone levels, T allele, beta 0.079 [0.059-0.099] unit decrease, p=7E-16, PMID 23408906]	Moderate
PDE10A	rs753760		C/C	chip	C (2 copy/s)	without judgment	Locus PDE10A of TSH [GWAS Catalog: Thyroid hormone levels, C allele, beta 0.100 [0.08-0.12] unit increase, p=1E-24, PMID 23408906]	Moderate
ITPK1	rs11624776		A/A	chip	A (2 copy/s)	without judgment	Locus ITPK1 of TSH [GWAS Catalog: Thyroid hormone levels, allele A, beta 0.064 [0.042-0.086] unit decrease, p=2E-9, PMID 23408906]	Moderate
TSHR	rs1991517		C/C	chip	G (0 copy/s)	without judgment	D727E (G allele): associated with slightly lower TSH; without clear clinical effect	Weak

Thyroid autoimmunity (Hashimoto and Graves)

Immune locks (PTPN22, CTLA4, SH2B3), HLA and the target genes (TSHR, TPO).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PTPN22	rs2476601	R620W	G/G	chip	A (0 copy/s)	favourable	R620W: the A allele weakens the T lymphocytes' brake; ~1.5-2 more Severe, Hashimoto, 1 type diabetes and rheumatoid arthritis	Strong
CTLA4	rs3087243	CT60	G/A	chip	G (1 copy/s)	unfavourable (light)	CT60: G allele reduces the soluble form of CTLA-4 (immune crossing); ~1.2-1.5x more Graves and Hashimoto	Strong
TSHR	rs179247		A/G	chip	A (1 copy/s)	unfavourable (light)	Intron 1 of TSHR: allele A reduces thymus receptor expression (less tolerance) and is associated with Graves disease	Strong
HLA-DR3/DQ2.5 (tag)	rs2187668		C/C	chip	T (0 copy/s)	favourable	Haplotype marker DR3-DQ2.5, the highest risk HLA for Graves, Hashimoto, diabetes type 1 and celiac	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SH2B3	rs3184504	R262W	C/C	chip	T (0 copy/s)	favourable	R262W: T allele is associated with more autoimmunity (hypothyroidism, celiac, type 1 diabetes)	Moderate
VAV3	rs4915077		T/T	chip	C (0 copy/s)	favourable	Locus VAV3 of hypothyroidism and autoimmunity of the thyroid [GWAS Catalog: Hypothyroidism, C allele, OR 1.30, p=8E-10, PMID 22493691]	Moderate
TPO	rs11675434		C/T	chip	T (1 copy/s)	unfavourable (light)	Locus TPO (thyroid peroxidase): anti-TPO antibodies [GWAS Catalog: Thyroid peroxidase antibody positivity, allele T, OR 1.21, p=2E-16, PMID 24586183]	Moderate
FLT3	rs76428106		T/T	chip	C (0 copy/s)	favourable	Intronic variant of FLT3 linked to hypothyroidism [GWAS Catalog: Hypothyroidism, C allele, beta 0.333 [0.29-0.37] unit increase, p=2E-65, PMID 41238958]	Moderate

Nodules and thyroid cancer

Loci of thyroid development (FOXE1, NKX2-1), growth (NRG1) and telomerase.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FOXE1 (9q22)	rs965513		A/G	chip	A (1 copy/s)	unfavourable (light)	Allele A reduces FOXE1 (thyroid development): ~1.7× more copy papillary cancer and also more hypothyroidism	Strong
NKX2-1/MBIP (14q13)	rs944289		C/T	chip	T (1 copy/s)	unfavourable (light)	The T allele reduces PTCSC3: ~1.3× more papillary cancer per copy	Strong
NRG1 (8p12)	rs2439302		G/G	chip	G (2 copy/s)	unfavourable	The G allele is associated with ~1.4× more papillary cancer per copy	Strong
DIRC3 (2q35)	rs966423		C/T	chip	C (1 copy/s)	unfavourable (light)	DIRC3 locus of thyroid cancer (and TSH) [GWAS Catalog: Thyroid cancer, C allele, OR 1.34, p=1E-9, PMID 22267200]	Moderate
PCNXL2 (1q42)	rs12129938		A/G	chip	A (1 copy/s)	unfavourable (light)	PCNXL2 locus of thyroid cancer [GWAS Catalog: Thyroid cancer, allele A, OR 1.32, p=4E-11, PMID 28195142]	Moderate
TERT	rs10069690		T/T	chip	T (2 copy/s)	unfavourable	Intron 4 of TERT: T allele is associated with longer telomeres and more thyroid and glioma cancer	Moderate
CHEK2	rs17879961	I157T	A/A	chip	G (0 copy/s)	favourable	I157T: CHEK2 reduced function; increased risk of thyroid cancer (~2×)	Strong

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 510-8$) of each set. The alleles are oriented towards more risk, or higher level in the sets of TSH and T4, and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
TSH (levels)	639 (586)	560 of 1172	583	3.9	without judgment
Free T4 (levels)	85 (78)	80 of 156	77	81.7	without judgment
Hypothyroidism and Hashimoto	1262 (1122)	1074 of 2244	1119	56.2	typical
Hyperthyroidism and Graves	261 (196)	151 of 392	196	50.1	typical
Goiter and nodules	264 (218)	200 of 436	218	89.7	unfavourable
Thyroid cancer	166 (137)	136 of 274	137	93.7	unfavourable

6. Favourable

- **Graves' disease:** genomic score at percentile 3.3 and 112 variant score in the 18.8 (33% coverage). **Moderate.**
- **Hypothyroidism and Hashimoto:** typical loci load (56.2) and typical scores (42.2, 38.2, 33.8, 51.8). **Moderate.**
- **PTPN22 R620W G/G**, HLA-DR3/DQ2.5 absent (rs2187668 C/C), **SH2B3 R262W C/C**, **VAV3 T/T** and **FLT3 T/T**: none of the thyroid autoimmunity risk alleles. Strong (PTPN22 and HLA).
- **DIO2 Thr92Ala T/T**: conversion of T4 into normal T3 in the brain and muscle. **Moderate.**
- **CHEK2 I157T absent** (A/A).

7. Unfavourable

- **Thyroid cancer:** percentiles 79.1, 82.5, 95.9 (41% coverage) and 80.7 (genomic); loci load at 93.7. **Moderate.**
- **Benign goiter:** percentiles 90.7 and 93.2 (genomic); loci load of goiter and nodules at 89.7. **Moderate.**
- **NRG1 rs2439302 G/G** and **TERT rs10069690 T/T** (2 copies each); **FOXE1 rs965513 A/G, NKX2-1 rs944289 C/T**, DIRC3 and PCNXL2 with 1 copy. Strong for FOXE1, NKX2-1 and NRG1 (each copy ~1.3-1.7×).
- **CTLA4 CT60 G/A, TPO C/T** and **TSHR rs179247 A/G**: 1 allele of slight risk of autoimmunity each, compensated by the rest of the profile. **Weak.**
- **Thermostat with low TSH and high free T4:** in mendelian randomization studies, a genetically lower TSH is associated with more goiter, nodules and thyroid cancer, and a higher free T4 is associated with more atrial fibrillation. **Weak to moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Thyroid alterations	Thermostat with low TSH (percentile 19.1) and high free T4 (percentile 80.1); nodular goiter above the 90th percentile	Unfavourable (mild)
Endocrine dysregulation	Stable pituitary–thyroid axis and low thyroid autoimmunity; Normal DIO2	Favourable
Somatic Mutation Accumulation	Papillary cancer appears by somatic mutations (BRAF, RAS) on a high polygenic background (FOXE1, NKX2-1, NRG1)	Unfavourable
Hayflick's Telomeres and Limit	TERT rs10069690 T/T: longer telomeres, at the expense of increased risk of thyroid cancer and glioma	Conditional
Chronic Inflammation	Without variants leading to autoimmune thyroiditis (PTPN22, HLA-DR3)	Favourable
Thermogenesis decline	Local T3 (DIO2) supports the thermogenesis of brown adipose tissue and muscle; preserved conversion	Favourable
Genomic instability	CHEK2 I157T absent; the risk of thyroid cancer is polygenic	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Thyroid panel in the next routine analysis:** TSH, free T4 and free T3, and anti-TPO antibodies as basal value. Interpreting with genetic thermostat: a low-normal TSH with normal free T4 and T3 is expected. If taking biotin, suspend 2–3 days before harvest, because it interferes with these dosages. *Suggestion (not applied).*
2. **Neck palpation in the annual consultation.** Screening ultrasound in people without symptoms is not recommended in current standards, because it discovers many microcarcinomas that would never cause problems. With goiter and thyroid cancer above the 80–the 95th percentile, a single basal ultrasound can be discussed with the physician; if normal, only signs are repeated. *Suggestion (not applied).*
3. **Signs that justify ultrasound at any time:** palpable nodule, increased volume of the neck, persistent hoarseness, difficulty in swallowing, increased cervical ganglion, or suppressed TSH with high free T4 (which, in this profile, suggests more an autonomous node than Graves). *Suggestion (not applied).*
4. **Adequate iodine, without excess:** about 150 µg/day from fish, shellfish, dairy or iodinated salt. The lack of iodine is the main environmental cause of goiter. Avoid algae or kelp supplements with high doses (> 500 µg/day), which can trigger dysfunction in a nodular thyroid. *Suggestion (not applied).*
5. **Food selenium** (fish, eggs, 1–2 chestnuts occasionally). There is no demonstrated benefit of selenium supplements outside the autoimmune disease, and excess is toxic. *Suggestion (not applied).*
6. **Saving the thyroid to ionizing radiation:** TC of the neck and thorax only when indicated, and thyroid protector on dental radiographs (see §94). Thyroid radiation is the strongest environmental risk factor of papillary cancer. *Suggestion (not applied).*
7. **Information:** with genetically high free T4, any treatment with thyroid hormone should be avoided without clear indication, because the excess of T4 increases the risk of atrial fibrillation and bone loss. *Suggestion (not applied).*

10. Limitations

- The goiter and thyroid cancer scores are based on health records. The lowest coverage (41% and 51%) should be read with caution; the genomic scores, with a coverage of 97%, point in the same direction.
- The chip does not test the rare variants of high effect of medullary thyroid cancer (RET) or family syndromes (PTEN, APC, DICER1). Only family history and a clinical panel exclude them.
- The loci of TSH and T4 explain only a part of the hormonal level; blood analysis always has priority over genetics.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. The alleles of the TSHR loci were manually verified in the literature.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (15 scores: 2 of hormonal levels, 4 of hypothyroidism and Hashimoto, 3 of hyperthyroidism and Graves, 2 of goiter and 4 of thyroid cancer). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** This study aimed to evaluate the effects of 2023 (TSH and T4), Teumer 2018 (thyroid function), Gudmundsson 2009 and 2012 (FOXE1, NKX2-1, NRG1), White 2025 and Gibson 2024 (scores), Brand 2009 (TSHR), Yuan 2020 (mendelian randomization TSH-cancer), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Thyroid thermostat (TSH, T4 and conversion to T3)	DIO2	rs225014	T/T	chip	C	0	strand +	without the risk allele
Thyroid thermostat (TSH, T4 and conversion to T3)	DIO1	rs2235544	C/A	chip	C	1	strand +	1 copy of effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	PDE8B	rs4704397	G/G	chip	A	0	strand +	0 copy of the effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	CAPZB	rs10917469	A/A	chip	A	2	strand +	2 copy of the effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	VEGFA	rs9472138	C/T	chip	T	1	strand +	1 copy of effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	PDE10A	rs753760	C/C	chip	C	2	strand + (palindromic; assumed +)	2 copy of the effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	ITPK1	rs11624776	A/A	chip	A	2	strand +	2 copy of the effect allele
Thyroid thermostat (TSH, T4 and conversion to T3)	TSHR	rs1991517	C/C	chip	G	0	without data	C/C: without the D727E variant; the TSH receptor is the usual
Thyroid autoimmunity (Hashimoto and Graves)	PTPN22	rs2476601	G/G	chip	A	0	strand +	without the risk allele
Thyroid autoimmunity (Hashimoto and Graves)	CTLA4	rs3087243	G/A	chip	G	1	strand +	1 copy of risk allele
Thyroid autoimmunity (Hashimoto and Graves)	TSHR	rs179247	A/G	chip	A	1	without data	A/G: 1 copy of TSHR intron A associated with Graves disease (moderate effect)
Thyroid autoimmunity (Hashimoto and Graves)	HLA-DR3/ DQ2.5 (tag)	rs2187668	C/C	chip	T	0	strand +	without the risk allele
Thyroid autoimmunity (Hashimoto and Graves)	SH2B3	rs3184504	C/C	chip	T	0	strand +	without the risk allele
Thyroid autoimmunity (Hashimoto and Graves)	VAV3	rs4915077	T/T	chip	C	0	strand +	without the risk allele
Thyroid autoimmunity (Hashimoto and Graves)	TPO	rs11675434	C/T	chip	T	1	strand +	1 copy of risk allele
Thyroid autoimmunity (Hashimoto and Graves)	FLT3	rs76428106	T/T	chip	C	0	strand +	without the risk allele
Nodules and thyroid cancer	FOXE1 (9q22)	rs965513	A/G	chip	A	1	strand +	1 copy of risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Nodules and thyroid cancer	NKX2-1/MBIP (14q13)	rs944289	C/T	chip	T	1	strand +	1 copy of risk allele
Nodules and thyroid cancer	NRG1 (8p12)	rs2439302	G/G	chip	G	2	strand + (palindromic; assumed +)	2 risk allele copies
Nodules and thyroid cancer	DIRC3 (2q35)	rs966423	C/T	chip	C	1	strand +	1 copy of risk allele
Nodules and thyroid cancer	PCNXL2 (1q42)	rs12129938	A/G	chip	A	1	strand +	1 copy of risk allele
Nodules and thyroid cancer	TERT	rs10069690	T/T	chip	T	2	strand +	2 risk allele copies
Nodules and thyroid cancer	CHEK2	rs17879961	A/A	chip	G	0	strand +	without the risk allele